

ADITYA BHARADE

+91-9899635879 adibharade@gmail.com [linkedin.com/in/AdityaBharade](https://www.linkedin.com/in/AdityaBharade) github.com/bharade

Education

Indian Institute of Technology Kharagpur

Nov. 2020 - June 2025

Dual B.Tech. (Hons.) and M.Tech.: Aerospace Engineering with Specialisation in AI, CGPA:8.84/10

Kharagpur, India

Birla Vidya Niketan

May 2020

Senior Secondary Examination(XII), 92.4%

New Delhi, India

Experience

IHI Corporation | AI Application Analyst

Oct. 2025 – Ongoing

Technology Planning Department, Corporate R&D

Yokohama, Tokyo, Japan

- Architected a **techno-economic simulation pipeline** in Python to benchmark SAF pathways and identify collaboration
- Identified manufacturing process optimizations yielding a **6.2% cost reduction** via automated sensitivity analysis
- Engineered a **time-series forecasting ensemble** (XGBoost, LightGBM) to model complex CH_4 reaction dynamics
- Achieved **98% prediction accuracy** on 10-minute horizons, enabling real-time anomaly detection for yield optimization
- Spearheading R&D on **Agentic CAD Workflows** to mitigate spatial hallucinations via structured **Chain-of-Thought**

École Polytechnique Fédérale de Lausanne (EPFL) | AI Researcher

Apr. 2025 – September 2025

Intelligent Maintenance and Operating Systems Lab | Guide: Prof. Olga Fink

Lausanne, Switzerland

- Developed a novel physics-informed **Graph Neural Network** pipeline for **fault localisation** using temporal Lamb waves
- Processed **71000+ time-series entries** in MATLAB and constructed fully connected graphs using PyTorch Geometric
- Enhanced generalisation by 35%** on unseen faults through a novel **attention-based spectral grid localiser**
- Implemented **DeepMind's SOTA Graph Network Simulator**, performing temporal rollout for increased adaptability
- Achieved an **additional 5% generalisation benefit** by benchmarking coupled model against traditional ML baselines

IHI Corporation | AI R&D Intern

May 2024 – July 2024

Hybrid Aircraft Program

Tokyo, Japan

- Developed a strategic roadmap for IT and business integration for the hybrid aircraft program over a 20 year timeline
- Demonstrated a **20% carrying cost reduction** and **50% stockouts reduction** using CNNs for inventory management
- Implemented a predictive maintenance tool using **XGBoost, LSTM and Autoencoders**, achieving a 0.89 F_1 -score
- Presented a clear **go-to-market strategy** for expanding decarbonisation operations worldwide and integrating modern AI

CSIR-National Aerospace Laboratories | Research Intern

May 2023 – June 2023

Computational Analyst | SU2, Gmsh, Python, MATLAB

Bengaluru, India

- Investigated the behaviour of transonic flow over launch vehicles upon cone angle and Mach number variation
- Deployed Stanford University's **SU2 solver based on C++** to study the pressure fluctuations over the launch vehicle
- Validated the wind tunnel results with numerical computations and observed a pair of counter rotating vortices in 3-D
- Processed the pressure coefficient distribution over the payload and boattail by using MATLAB and Python
- Observed a **31.2% increase in pressure coefficient** on decreasing cone angle from 15° to 25° at a 0.94 Mach number

Publications

Strategic Design Approach for Fuselage BLI Engine Based Hybrid Propulsion Engine

Oct. 2023

8th Gas Turbine India Conference, ASME International Gas Turbine Institute

Bengaluru, India

Projects

Master's Thesis: Transformer-Based Damage Diagnosis in CFRP | Python, MATLAB

Aug. 2024 – Dec. 2024

Guide: Dr. Mira Mitra, Assistant Professor, Department of Aerospace Engineering, IIT Kharagpur

- Deployed a **transformer architecture based classifier** to detect structural damage from time-series wave responses
- Designed a **scalable pipeline** for data ingestion, augmentation, model training, and evaluation using TensorFlow
- Achieved an **85.57%** and **84.14%** accuracy on training data and test data from robust confusion matrices
- Evaluated model performance using robust metrics observing **precision, recall and F_1 -score of 0.91, 0.74, 0.82**

Application of Transfer Learning in Structural Health Monitoring | Python, MATLAB

Jan. 2024 – May 2024

Guide: Dr. Mira Mitra, Assistant Professor, Department of Aerospace Engineering, IIT Kharagpur

- Implemented a **transfer learning** based method for rapid adaptation and **software reusability** for damage diagnosis
- Preprocessed the guided signals at using MATLAB and conducted **zone-wise analysis** for computational efficiency
- Trained a **convolutional autoencoder** for feature extraction with a **97.81% accuracy** for reconstructed wave signals
- Transferred the feature extractor for damage diagnosis, achieving a **76.25% target accuracy and 0.76 precision**

Scalable Text Summarization System | Python, Transformers, Docker, FastAPI

Oct. 2024 – Nov. 2024

- Developed an end-to-end text summarisation application using a **Google Pegasus transformer** from Hugging Face
- Designed and deployed a **data processing pipeline** for text cleaning, tokenization, and feature extraction, using FastAPI
- Established a **CI/CD pipeline** with **GitHub Actions** to automate build, test, and deployment process

Awards

Honorable Mention | Technology Excellence | IIT Kharagpur **May 2025**

- **Recognised among 15000+ students** for outstanding contributions and innovations in technology and research acumen

Honorable Mention | Badminton | IIT Kharagpur **May 2025**

- Felicitated for outstanding achievements at the nationwide Inter IIT Sports Meet(2023-24) and intra-university events

Competitions

3rd Position | Engine Design Competition | AIAA **Nov. 2022 – June 2023**

Hybrid Boundary Layer Ingestion Turbofan *San Diego, California*

- Designed a **hybrid turbofan** based on the CFM56-7B24 with boundary layer ingestion for the NASA Starc-ABL aircraft
- Replicated the baseline engine using GasTurb 14. Spearheaded the materials research, intake design and nozzle design
- Met the thrust requirement with a **fuel benefit of 13.06%** at the design point with a reduction in overall length
- Secured a **top 3 finish worldwide**; nominated to present before the AIAA forum in San Diego, California

Gold | Inter IIT Tech Meet 11.0 | Indian Institute of Technology Kanpur **Feb. 2023**

Engineer's Conclave *Kanpur, India*

- Achieved a **perfect score** and the overall **Gold among 20+ IITs** for applications of Rapid Expansion Fans in aviation
- Explored the role of expansion fans in jet deflection for thrust vectoring, rapid cooling and hypersonic flow pockets
- Conducted a series of experimental studies and visualised the same using numerical and experimental schlieren imaging

Relevant Coursework

- **AI/ML:**Machine Learning: Foundations and Applications | Advanced Learning Paradigms in AI | Dependable and Secure AI/ML | Artificial Intelligence: Foundations and Applications | Graphical and Generative Models for Machine Learning
- **Mathematics/CSE:** Programming and Data Structures | Numerical Solutions of Ordinary and Partial Differential Equations | Linear and Numerical Analysis | Transform Calculus
- **Engineering:** Flight Vehicle Controls | Aircraft Stability and Controls | Mechanics of Flight | Rocket Propulsion | Automatic Control of Aircraft | Advance Gas Turbine Theory | Aerospace Propulsion Systems | Space Dynamics | Low and High Speed Aerodynamics | Aerospace Structural Analysis and Dynamics | Unsteady Aerodynamics | Smart Structures

Technical Skills

Languages/Frameworks: C, C++(STL), Python, HTML, CSS, Git

Libraries: HuggingFace, TensorFlow, PyTorch, NumPy, Pandas, Transformers, Keras, Sci-kit Learn

Softwares and Tools: SU2, Gmsh, Paraview, Linux, MATLAB, ANSYS, SolidWorks, GasTurb, Jupyter, MLFlow

Positions of Responsibility

Mentor | Department of Aerospace Engineering | IIT Kharagpur **Oct. 2021 – Dec. 2022**

- Mentored 10 undergraduate freshmen in academic and extra-curricular aspects as a part of the Student's Welfare Group

Extracurricular

- **Badminton:** Represented IIT Kharagpur at the 55th Inter IIT Sports Meet(2022) held at IIT Delhi, Bronze: Udghosh'22-The annual sports fest held at IIT Kanpur; Zonals: **Gold:** Under-19(2017),Under-15(2016);**Silver:** Under-17(2018);**Bronze:** Under-19(2019); Participated in the National Central Zone tournament organized by CBSE
- **Linguistic:** Pursued French from Grade 6 to Grade 10. Achieved **International rank 11** in the International French Language Olympiad organized by the French embassy (2018)